Shobin Sam S

AI/ML Engineer | Generative AI | RAG & Agentic Systems
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SUMMARY

AI/ML Engineer with 4+ years of experience in machine learning and Generative AI. Skilled in building LLM applications, agentic workflows, RAG pipelines, and knowledge graph solutions, with strong Python expertise and hands-on experience deploying ML systems to production.

EDUCATION

Amrita College of Engineering and Technology — **Nagercoil, Tamil Nadu, India**
Bachelor of Engineering in Computer Science — *Sep 2018 – Jul 2022*

TECHNICAL SKILLS

Languages: Python, SQL (PostgreSQL)
Frameworks & Libraries: PyTorch, LangChain, LangGraph, Hugging Face Transformers, Sentence Transformers, Pandas, NumPy, OpenCV, scikit-learn, Matplotlib
Machine Learning & Generative AI: Deep Learning, NLP, LLMs, RAG, Agent Orchestration, Tool Calling, Model Context Protocol (MCP), Embeddings, Hybrid Retrieval, Cross-Encoder Reranking, Knowledge Graphs
Data & Retrieval: Qdrant, FAISS, Neo4j, Redis
Deployment & Tools: FastAPI, Streamlit, Ollama, Celery, Docker Compose, Git, GitHub Actions, OpenTelemetry, Prometheus, Grafana

WORK EXPERIENCE

AI/ML Developer
Tata Consultancy Services, Trivandrum, India — *Aug 2022 – Present*

- Developed **LLM agents** for a **ticketing tool and company FAQ chatbot** under active development, supporting TCS workflows handling **15,000–28,000 tickets per quarter** through **knowledge graphs, tool calling, and MCP integrations**.
- Developed an **automated document validation system** using **YOLO, Azure Computer Vision, and Azure Document Intelligence**, achieving **98% PoC accuracy** and **89% production accuracy**, and reducing manual verification effort by **approximately 50%**.
- Led **end-to-end production deployment** of the document validation ML product, coordinating cross-functional teams to meet a **one-month delivery timeline**.
- Built an **IoT-based water quality prediction system** using Raspberry Pi and sensor data to predict bacterial presence with **82% accuracy**; developed cloud-based ML pipelines and contributed to a **patent-pending prediction approach**.
- Built a **Generative AI email drafting and evaluation tool** using **LLaMA 7B and Streamlit**, with real-time quality scores and suggestions to improve email effectiveness.
- Delivered **AI/ML solution presentations** to senior leadership through internal innovation programs, with several ideas recognized as best solutions.

PROJECTS

CortexOps AI – Agentic AIOps Investigation Platform

- Built an AIOps tool using **FastAPI, LangGraph, and Ollama** to investigate model-accuracy drops by comparing metrics, deployments, code changes, and logs. GitHub
- Integrated **Qdrant hybrid RAG, reranking, and knowledge graphs** with **five MCP servers** to produce reports with source citations; implemented resumable agent workflows and approval before issue creation.

Hand Gesture Calculator

- Built a real-time calculator using **YOLO** for hand gesture detection and **Streamlit** for the interface; mapped detected gestures to arithmetic operations. GitHub

CERTIFICATIONS

- Claude Certified Developer – Foundations | Anthropic Certificate — Sep 2026
- Microsoft Certified: Azure AI Engineer Associate Certificate — 2025